Frequency-Domain Compensator Design

Designing lead/lag compensators straight from the Bode plot.

Ogata, *Modern Control Engineering*, Ch. 7–8.

In this tutorial you will:

- size a lead compensator from a phase-margin specification,
- place its corner frequencies and recompute the margin, and
- see the lag compensator as the magnitude-attenuation dual.

Step through with **Ctrl+Enter**, or render a report with `publish`.

This file parallels the root-locus compensator designs in `Root-Locus/LeadCompensation.m` and `Root-Locus/LagCompensation.m`, but works directly with phase margin and Bode plots instead of pole/zero geometry in the s-plane.

Contents

Lead Compensator: Phase-Margin Specification

A phase-lead compensator $G_c(s) = K_c \frac{1+\alpha T s}{1+Ts}$, $\alpha > 1$, adds positive phase. The maximum phase added,

$\phi_{max} = \sin^{-1} \frac{\alpha-1}{\alpha+1}$

occurs at $\omega_m = \frac{1}{T\sqrt{\alpha}}$, where the magnitude contribution is $10 \log_{10} \alpha$ dB. Design procedure:

1. Find K to satisfy any steady-state error spec; plot the uncompensated Bode diagram.
2. Measure the uncompensated phase margin PM_1 at the gain crossover.
3. Compute the required additional phase $\phi_{max} = PM_{desired} - PM_1 + \epsilon$ (extra $\epsilon \approx$ 5-12° to allow for the crossover frequency shift).
4. Solve for $\alpha = \frac{1+\sin \phi_{max}}{1-\sin \phi_{max}}$.
5. Find the new crossover ω_m where the uncompensated magnitude equals $-10 \log_{10} \alpha$ dB; set $T = \frac{1}{\omega_m \sqrt{\alpha}}$.

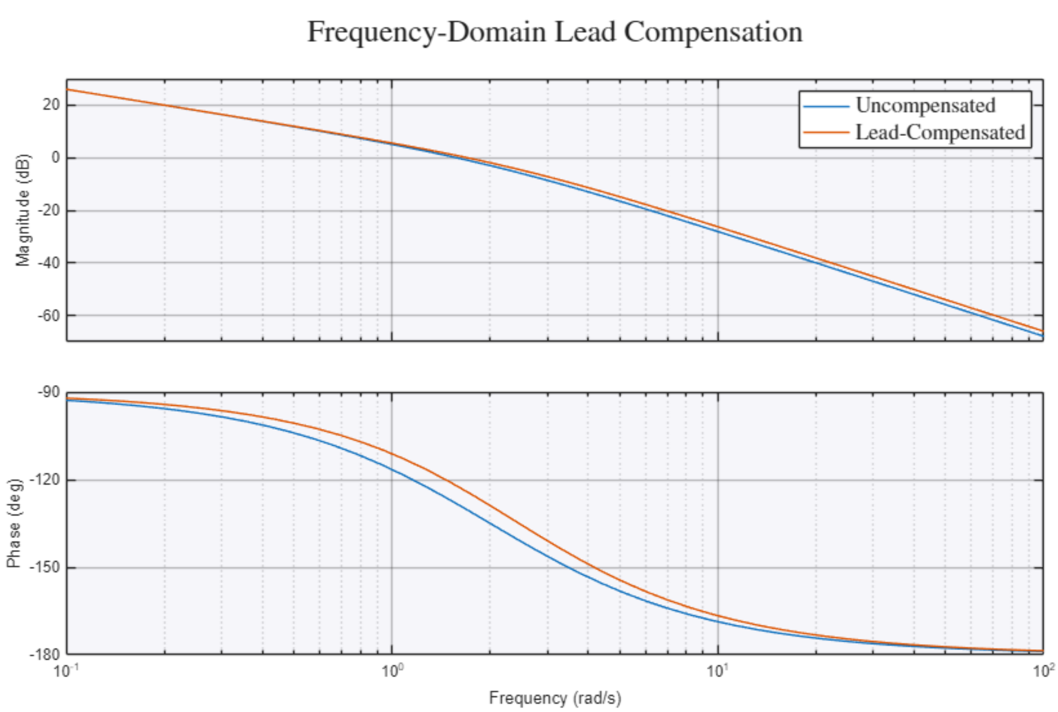
Worked Example

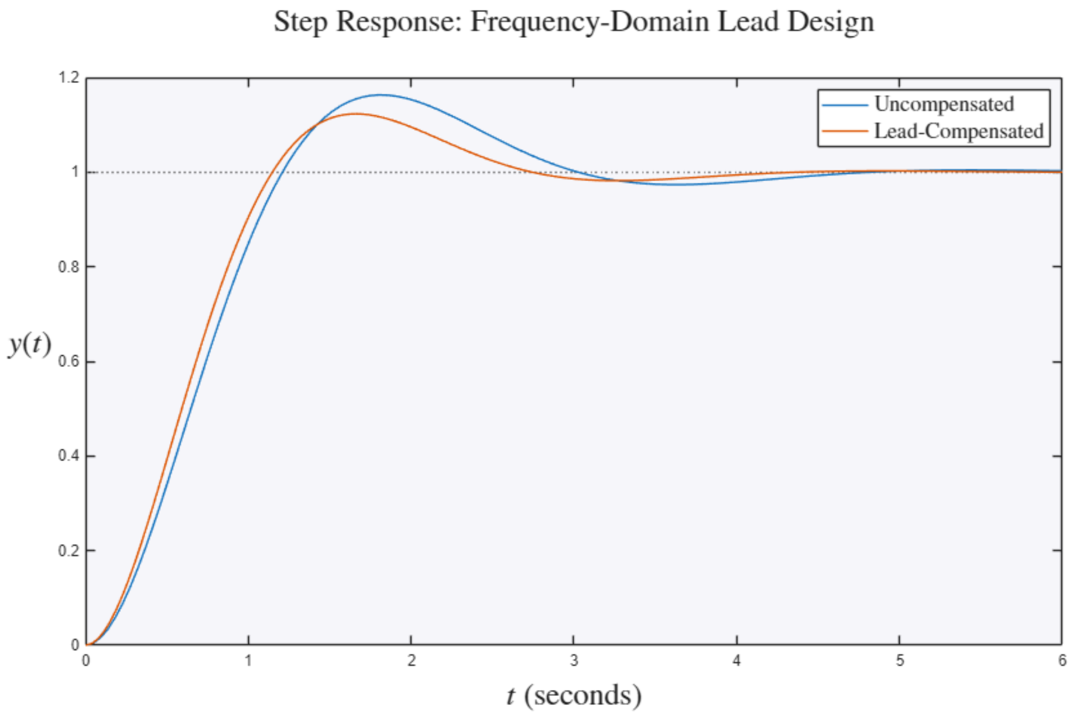
$G(s) = \frac{4}{s(s+2)}$, design spec: $PM \geq 50°$.

```
G = tf(4,[1 2 0]);

PM_desired = 50;
[~,PM1,~,wc1] = margin(G);
fprintf('Uncompensated PM = %.2f deg at wc = %.2f rad/s\n', PM1, wc1)

epsilon = 8;  % extra phase margin to compensate for crossover shift
phi_max = (PM_desired - PM1 + epsilon)*pi/180;
alpha = (1+sin(phi_max))/(1-sin(phi_max));
fprintf('Required phi_max = %.2f deg, alpha = %.4f\n', phi_max*180/pi, alpha)

% Magnitude needed from compensator at wm: -10*log10(alpha) dB.
% Find wm where |G(jw)| = -10*log10(alpha) dB.
mag_needed_dB = -10*log10(alpha);
w_sweep = logspace(-1,2,2000);
```

```
[mag,~] = bode(G,w_sweep);
mag_dB = 20*log10(squeeze(mag));
[~,idx] = min(abs(mag_dB - mag_needed_dB));
wm = w_sweep(idx);
T_lead = 1/(wm*sqrt(alpha));
fprintf('wm = %.4f rad/s, T = %.4f\n', wm, T_lead)

Gc = tf([alpha*T_lead 1],[T_lead 1]);
Kc = 1/abs(evalfr(Gc,1j*0))*1;  % unity DC gain on the compensator network itself
Gc_full = Kc*Gc;

G_comp = series(Gc_full,G);
[~,PM_comp,~,wc_comp] = margin(G_comp);
fprintf('Compensated PM = %.2f deg at wc = %.2f rad/s\n', PM_comp, wc_comp)

figure
hold on
bode(G)
bode(G_comp)
hold off
legend('Uncompensated','Lead-Compensated','Interpreter','latex','FontSize',14)
title('Frequency-Domain Lead Compensation','Interpreter','latex','FontSize',20)
grid on

T_uncomp = feedback(G,1);
T_comp = feedback(G_comp,1);
figure
hold on
step(T_uncomp)
step(T_comp)
hold off
legend('Uncompensated','Lead-Compensated','Interpreter','latex','FontSize',14)
title('Step Response: Frequency-Domain Lead Design','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

```
Uncompensated PM = 51.83 deg at wc = 1.57 rad/s
Required phi_max = 6.17 deg, alpha = 1.2410
wm = 1.7007 rad/s, T = 0.5278
Compensated PM = 55.84 deg at wc = 1.70 rad/s
```





Lag Compensator: Phase-Margin Specification (Brief Note)

A phase-lag compensator $G_c(s) = K_c \frac{1+Ts}{1+\beta Ts}$, $\beta > 1$, is placed so its corner frequencies sit roughly a decade below the new gain crossover, leaving phase nearly unaffected there while attenuating magnitude by $20 \log_{10} \beta$ dB to pull the crossover down to a frequency where the existing phase already meets the margin spec – the frequency-domain dual of `Root-Locus/LagCompensation.m`, which pursued the same goal (steady-state error improvement without disturbing the dominant transient response) via pole/zero placement instead of a magnitude-attenuation argument.

Try it yourself

- Ask for a larger phase margin (`PM_desired = 65`) and notice the lead ratio `alpha` grow – more phase means a wider pole/zero split.
- Reduce the safety margin `epsilon` to 3 and see the achieved margin fall short (the crossover shifts more than a small epsilon allows for).